

Project: Volatility, VaR, Expected Shortfall & Portfolio Optimization

Cuauhtemoc Daniel Puente Cavazos

```
## Loading required package: xts

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric

## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.4.3

Load packages
```

```
options(repos = c(CRAN="http://cran.rstudio.com"))
if (!require("MASS")) install.packages("MASS")
if (!require("forecast")) install.packages("forecast")
if (!require("rugarch")) install.packages("rugarch")
if (!require("rmgarch")) install.packages("rmgarch")
if (!require("timetk")) install.packages("timetk")
if (!require("quantmod")) install.packages("quantmod")
library(forecast); library(MASS); library(rugarch); library(rmgarch); library(quantmod); library(graphics)
library(xts); library(timetk); library(stats); library(ggplot2); library(ggfortify); library(ggpubr)
```

1. Univariate Volatility

1.1 Data

S&P500 index ([Source](#)), for plots see file “9a. Volatility”

```
NASDAQ <- quantmod::getSymbols("^IXIC", src="yahoo", return.class="xts", from="2010-01-01", auto.assign=FALSE)
```

```
#print(colnames(NASDAQ))
NASDAQ_close <- NASDAQ[, 4]
colnames(NASDAQ_close) <- "NASDAQ"
NASDAQ_r <- na.omit(diff(log(NASDAQ_close))) * 100
tail(NASDAQ_close)
```

```
##              NASDAQ
## 2026-05-27 26674.73
## 2026-05-28 26917.47
## 2026-05-29 26972.62
## 2026-06-01 27086.81
## 2026-06-02 27093.90
## 2026-06-03 26853.98
```

1.2 Univariate Volatility Model

ARMA(1,0)-GARCH(1,1) model specification

```
model <- rugarch::ugarchspec(  
  variance.model = list(model="sGARCH", garchOrder=c(1, 1)),  
  mean.model = list(armaOrder=c(1, 0)))
```

model fit

```
model_fit <- rugarch::ugarchfit(spec=model, data=NASDAQ_r, solver='hybrid')
```

Forecast

```
n_future <- 8  
model_forecast <- rugarch::ugarchforecast(fit=model_fit, n.ahead=n_future)  
mean_f <- as.numeric(model_forecast@forecast$seriesFor)/100 # from % to fraction  
std_f <- as.numeric(model_forecast@forecast$sigmaFor)/100 # from % to fraction
```

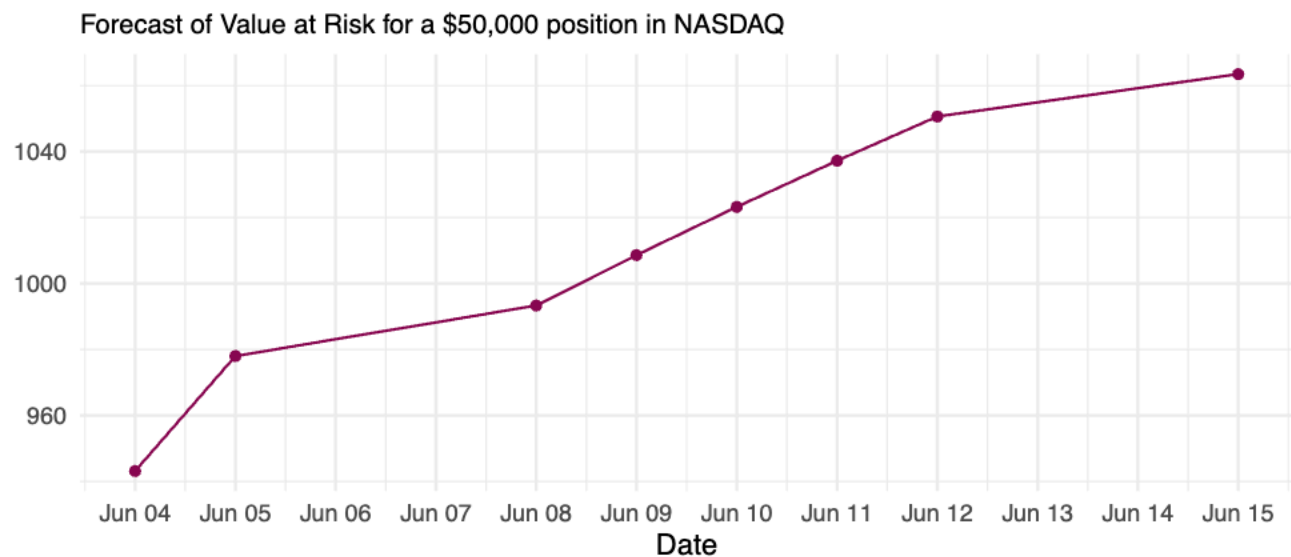
1.3 Value at Risk

At $\alpha = 1\%$ for a \$50,000 position in NASDAQ

```
VaR_f <- 50000 * abs(mean_f - 2.33 * std_f)
```

Plot

```
id <- timetk::tk_index(NASDAQ_r)  
id_f <- timetk::tk_make_future_timeseries(id, length_out=n_future, inspect_weekdays=TRUE)  
  
VaR_f <- xts(VaR_f, order.by=id_f)  
colnames(VaR_f) <- "VaR"  
  
ggplot(data=VaR_f, aes(x=index(VaR_f), y=VaR)) +  
  geom_line(color="deeppink4") +  
  geom_point(color="deeppink4") +  
  labs(x="Date", y="", title = "Forecast of Value at Risk for a $50,000 position in NASDAQ") +  
  theme_minimal() + scale_x_date(date_breaks="1 day", date_labels = "%b %d") +  
  theme(plot.title = element_text(size=10))
```



1.4 Expected Shortfall

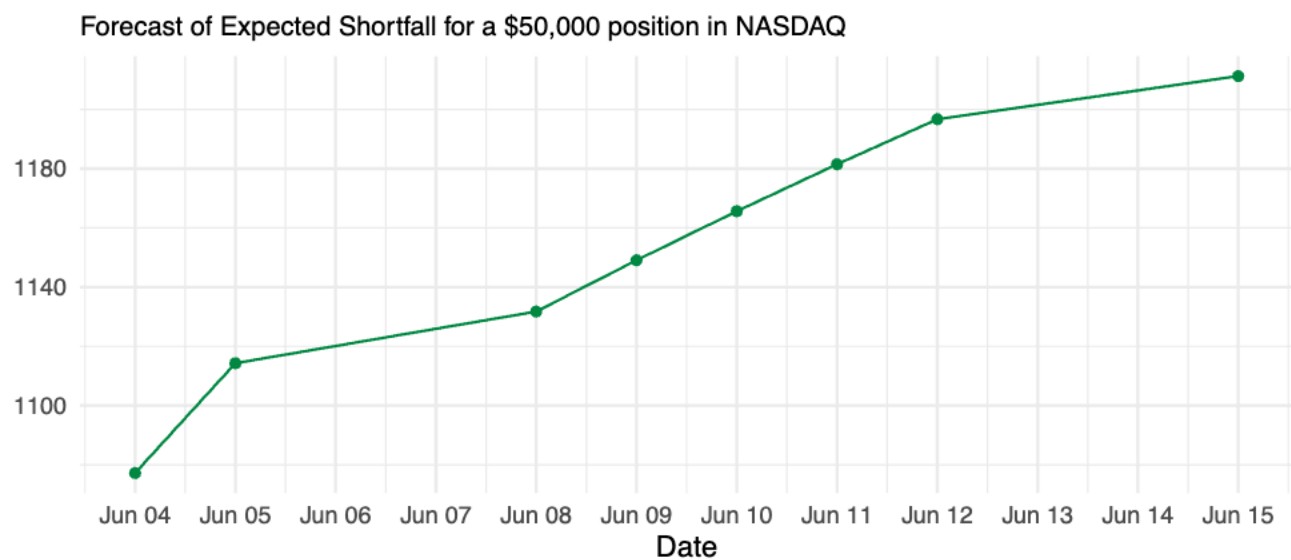
At $\alpha = 1\%$ for a \$50,000 position in NASDAQ

```
ES_f <- 50000 * abs(mean_f - 2.64 * std_f)
```

Plot

```
ES_f <- xts(ES_f, order.by=id_f)
colnames(ES_f) <- "ES"

ggplot(data=ES_f, aes(x=index(ES_f), y=ES)) +
  geom_line(color="springgreen4") +
  geom_point(color="springgreen4") +
  labs(x="Date", y="", title = "Forecast of Expected Shortfall for a $50,000 position in NASDAQ") +
  theme_minimal() + scale_x_date(date_breaks="1 day", date_labels = "%b %d") +
  theme(plot.title = element_text(size=10))
```



2. Multivariate Volatility

2.1 Data

These come from, [tickers](#)

```
names <- c("Pfizer", "AstraZeneca", "Johnson&Johnson", "Merck", "Eli Lilly", "Bristol-Myers Squibb")
tickers <- c("PFE", "AZN", "JNJ", "MRK", "LLY", "BMY")

color_palette <- c("lightblue3", "darkorange", "yellowgreen", "cyan3")

findat <- quantmod::getSymbols(tickers, src="yahoo", from="2022-06-01")
prices_lst <- lapply(tickers, function(ticker) {
  get(ticker)[, paste0(ticker, ".Close")]
})
prices <- do.call(merge.xts, prices_lst)
colnames(prices) <- tickers
returns <- na.omit(diff(log(prices)))*100 # in % terms
```

Risk-free asset return, U.S. 3-month Treasury Bills ([Source](#))

```
rf_rate <- 4.5 / 252
```

2.2 Multivariate Volatility Model

DCC model

```
# Univariate GARCH model specification
uni_garch <- rugarch::ugarchspec(
  variance.model = list(model = 'sGARCH', garchOrder = c(1,1)),
  mean.model = list(armaOrder = c(1,1)))

# Replicate the univariate model for each dimension
dim <- ncol(returns)
rep_garch <- rugarch::multispec(replicate(dim, uni_garch))

# Formulate the DCC model
DCC_model <- rmrgarch::dccspec(uspec = rep_garch, dccOrder = c(1, 1), model = "DCC")

# Fit the model to the data
DCC_fit <- rmrgarch::dccfit(spec = DCC_model, data = returns, solver = c("hybrid", "solnp"))
```

```
## Warning in arima(data, order = c(modelinc[2], 0, modelinc[3]), include.mean =
## modelinc[1], : possible convergence problem: optim gave code = 1
```

forecast of the Covariance matrix, 1 step ahead

```
fct <- rmrgarch::dccforecast(DCC_fit, n.ahead=1)
S <- fct@mfcast$H[[1]]
S <- as.matrix(S[, , 1])
S
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## [1,] 2.0297872 0.8647031 0.5627555 0.8568244 0.9464393 1.1657790
## [2,] 0.8647031 2.1884086 0.6144245 0.9963572 1.1719328 0.9505286
## [3,] 0.5627555 0.6144245 1.2089778 0.8845885 0.8298689 0.7806058
## [4,] 0.8568244 0.9963572 0.8845885 2.7481972 1.3582567 0.9534049
## [5,] 0.9464393 1.1719328 0.8298689 1.3582567 5.5353477 1.2386502
## [6,] 1.1657790 0.9505286 0.7806058 0.9534049 1.2386502 2.8082966
```

Invert the forecasted Covariance matrix

```
S_inv <- MASS::ginv(S)
S_inv
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## [1,] 0.72689511 -0.13415917 -0.07029640 -0.07451993 -0.02188962 -0.20184544
## [2,] -0.13415917 0.65013801 -0.10278919 -0.10707526 -0.05633145 -0.07459182
## [3,] -0.07029640 -0.10278919 1.24685716 -0.25678672 -0.05148945 -0.17272060
## [4,] -0.07451993 -0.10707526 -0.25678672 0.54306343 -0.05447154 -0.02178798
## [5,] -0.02188962 -0.05633145 -0.05148945 -0.05447154 0.22608524 -0.03876051
## [6,] -0.20184544 -0.07459182 -0.17272060 -0.02178798 -0.03876051 0.53762804
```

forecast of the mean, 1 step ahead

```
mu <- fct@mforecast$mu
mu <- as.vector(mu)
mu
```

```
## [1] -0.01615857 0.02285934 0.03181073 0.02307786 0.18020122 -0.07989684
```

2.3 Optimal Tangency Portfolio Weights

```
mu <- as.numeric(mu)

x_star <- (S_inv %*% (mu - rf_rate)) / sum(S_inv %*% (mu - rf_rate))
x_star
```

```
##           [,1]
## [1,] 19.288339
## [2,] -7.229338
## [3,] -48.195910
## [4,] 9.948872
## [5,] -72.773079
## [6,] 99.961116
```

```
x_star_pct <- x_star * 100
for (i in seq_along(names)) {
  cat(sprintf("%-24s: %.2f%%\n", names[i], x_star_pct[i]))
}
```

```
## Pfizer           : 1928.83%
## AstraZeneca      : -722.93%
## Johnson&Johnson : -4819.59%
## Merck            : 994.89%
## Eli Lilly        : -7277.31%
## Bristol-Myers Squibb : 9996.11%
```